Name: ____________

Date: ____________

Power Functions & Function Comparison

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can identify a power function and interpret the parameters in $y = kx^p$.**

RELATED SKILLS

- I — Interpret the coefficient and exponent in a power function.
- I — Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
- I — Represent and interpret a power-variation relationship.

☐ **2. I can construct a power function from two data points or a proportional relationship.**

RELATED SKILLS

- I — Determine a power function from two points or a proportional relationship.
- A — Isolate a specified variable in an equation or formula.

☐ **3. I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.**

RELATED SKILLS

- I — Determine power-function behavior as the input approaches zero or positive or negative infinity.
- I — Relate the coefficient and exponent of a power function to its graph shape.
- D — Connect parameters in a formula to features of its graph.

☐ **4. I can recognize and model direct, inverse, and other power-variation relationships.**

RELATED SKILLS

- I — Represent and interpret a direct-variation relationship.
- I — Represent and interpret an inverse-variation relationship.
- D — Represent and interpret a power-variation relationship.

☐ **5. I can solve equations involving integer or rational powers.**

RELATED SKILLS

- A — Interpret numerator and denominator roles in a rational exponent.
- A — Use inverse operations to maintain an equivalent equation.

☐ **6. I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.**

RELATED SKILLS

- I — Compare the long-run growth or decay of functions from different families.
- D — Compare functions shown in different representations.

☐ **7. I can fit a power regression model and interpret its parameters and correlation evidence.**

RELATED SKILLS

- I — Use technology to determine a power regression model.
- D — Interpret the coefficient and exponent in a power function.
- A — Interpret the sign and magnitude of a correlation coefficient in context.